

Fig. 2: An actual-size, single-side PCB for the loudspeaker protection circuit

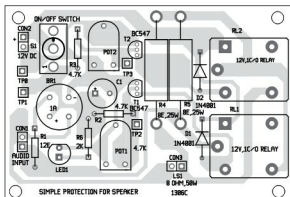


Fig. 3: Component layout for the PCB

If the input signal is higher than the threshold defined by POT1 and lower than the threshold defined by POT2, only R5 will be applied in series with the loudspeaker. When the input signal is higher than both the thresholds, all N/C contacts are open and loudspeaker LS1 is connected to connector CON1 through protection resistors R4 and R5.

Construction and testing

An actual-size, single-side PCB for the loudspeaker protection circuit is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on a PCB, enclose it in a suitable case. Fix CON1 and CON2 on the

front side of the case for connecting the audio power amplifier and the 12V battery/adaptor. Fix CON3 on the rear side of the case for connecting the speakers you want to protect.

To test and configure the circuit, apply 12V power supply at CON2 and switch S1 'on'. Take care that no audio signal is applied. Check the voltage at TP1 with respect to TP0 as shown in the table. Set low and high thresholds using POT1 and POT2. The same can be checked at TP2 and TP3, respectively. It is preferable to test and adjust the circuit with appropriate sine wave generator applied to CON1.

If not available you can close the switch S1 and use the power supply of the unit in order to test the activation of the relays and set the threshold points of the trimmers. ●

The author is a regular contributor to EFY. He has worked as a researcher and assistant professor in Technical University of Sofia (Bulgaria) and as a lecturer in Kingdom of Morocco. He is now an electronics engineer in the private sector in Bulgaria.

EFY BOOKS
SERIES

Books For You!



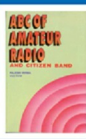
Rs 180/- (with CD)

Working principles, common faults and remedies, plus simple testing procedures.



Rs. 60/-.

By D. Venkatasubbiah
clears many misconceptions
and doubts about small
batteries and cells



Rs 75/-

Explains basics of electronics besides Ham radio and Citizen Band



Rs 804

By Gp. Capt. (Retd) K.C. Bhasin
Instructions for 40 popular
and tested electronic hobby
projects, plus much more



Rs. 300/- (with CD)

An application-oriented and resourceful book for engineering students, hobbyists, teachers and practicing engineers alike.



Rs. 300/-

Collection of 60 simple projects, highly useful for engineering students, Teachers & Hobbyists

Distributor: IBH Books & Magazine Distributors (P) Ltd

Ahmedabad
079-26131261, 26052750
Chandigarh
0172-2703667
Kolkata
033-22498096, 22498097

Bengaluru
080-22261305, 22252385
Chennai
044-28592743, 28592562
Pune
020-26131261, 26052756

Mumbai
022-40497413, 40497474
Hyderabad
040-24750053
Thiruvananthapuram
0471-2475443

New Delhi
011-23313014/15
Patna
0612-2226540
Jaipur
0141-2317661

For postage-free Delivery:

Kits'n'Spares
D-88/5, Okhla Industrial Area, Phase 1,
New Delhi 110020; Phone: 011-26371661/62
E-Mail: info@kitsnspares.com
Website: www.kitsnspares.com